

EEG Pain Biomarker Pipeline

Reproducible Preprocessing · Feature Engineering · Leakage-Safe LOSO Validation

OpenNeuro ds005284 · 26 Biosemi subjects · C3 / CZ / C4 somatosensory ROI

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Goals & Motivation

Why leakage-safe EEG pain biomarkers matter

02

Dataset & Provenance

ds005284 facts, epoch counts, ROI

03

Preprocessing Pipeline

Exact reproducible steps from raw BDF

04

Feature Families (7)

Spectral → Nonlinear → Connectivity

05

Key Biomarker Findings

CZ complexity + low-gamma + multiband

06

LOSO Validation Design

Train-only scaling, selection, metrics

07

Classifiers & Ablations

LDA deployment vs ensemble exploration

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Outputs, Audit & Checklist

Artifacts + new-dataset playbook

1. Robust Biomarkers

Identify EEG features that reliably discriminate pain vs no-pain states across subjects (not just within).

2. Leakage-Safe Validation

Leave-One-Subject-Out (LOSO) with all scaling & selection fit exclusively on training folds.

3. Deployable & Auditable

Interpretable model (LDA preferred)
full config JSON + fold audit CSVs
ready for real-time or clinical QC.

Design principle

Every statistic used for scaling, filtering or feature ranking is computed on the training subjects only.
This prevents optimistic bias that is common in EEG machine-learning pipelines.

OpenNeuro ds005284 (Zhao et al.)

Acquisition	Biosemi 64-channel, 1024 Hz
Resampled to	256 Hz (pipeline standard)
Subjects	26 healthy volunteers
Epochs total	781 (367 pain / 414 no-pain)
ROI electrodes	C3, CZ, C4 (S1 focus)
Stimulus	Laser-evoked pain (condition 54)
License	CC0 — fully open
Reference	Average reference after filter

Epoch Windows

- Pain: 0 → +2 s post-onset
- No-pain baseline: −5 → −3 s
- Aligned to laser stimulus

Artifact Rule

- Reject |amplitude| > 200 μ V
- Optional SNR / flat-channel QC
- Per-subject QC log kept

Why this ROI?

- Primary somatosensory cortex
- Strong laser-evoked responses
- Scalable to other S1 studies

1

Channel map

Biosemi labels → 10–20 via channels.tsv

2

Resample

1024 Hz → 256 Hz (scipy.signal.resample)

3

Notch

50 Hz & 100 Hz (or 60/120 for US)

4

Bandpass

0.5 – 80 Hz Butterworth IIR

5

Reference

Average reference (post-filter)

6

Epoch

Pain 0→+2 s | Baseline –5→–3 s

7

Reject

Amplitude $|\mu V| > 200$ (dataset-tuned)

8

QC save

Per-epoch flags + subject_summary.csv

Spectral

Abs / rel band power
 δ θ α β low- γ
Ratios, centroid, 1/f

Nonlinear

Higuchi & Petrosian FD
DFA, SampEn, PermEn
Lempel-Ziv, Hjorth

Time-domain

RMS, STD, peak-to-peak
Line length, ZCR
Skew, kurtosis, crest

Wavelet

db4 energy ratios
Approx / detail bands
Wavelet entropy

MFCC

First 5 MFCC means
Per ROI channel
librosa backend

AR Dynamics

Burg AR coeffs
Innovation variance
Low-order model

Connectivity

Coherence per band
PLV (C3-CZ, C3-C4,
CZ-C4) pairs

■ Band power & spectral

abs/rel δ θ α β low- γ (30–45 Hz), log powers, α/β & γ/α ratios, spectral centroid / median / edge, α -peak, 1/f slope

■ Nonlinear & complexity

Higuchi FD, Petrosian FD, DFA, Sample Entropy, Permutation Entropy, Spectral Entropy, Lempel-Ziv, Hjorth (activity / mobility / complexity)

■ Time-domain statistics

RMS, standard deviation, peak-to-peak, line length, zero-crossing rate, crest factor, skewness, kurtosis

■ Wavelet (PyWavelets)

db4 decomposition energy ratios (approx & detail), wavelet entropy

■ MFCC (librosa)

Mean of first 5 Mel-frequency cepstral coefficients per channel

■ Autoregressive

Burg AR coefficients (low order) + innovation variance

■ Connectivity

Band-wise coherence & Phase-Locking Value for pairs C3-CZ, C3-C4, CZ-C4

Best single-channel

Nonlinear complexity at CZ

Higuchi FD, DFA,
SampEn / PermEn,
Petrosian FD

Best single band

Low-gamma (30–45 Hz)

Still weaker than
multiband multi-feature
combinations

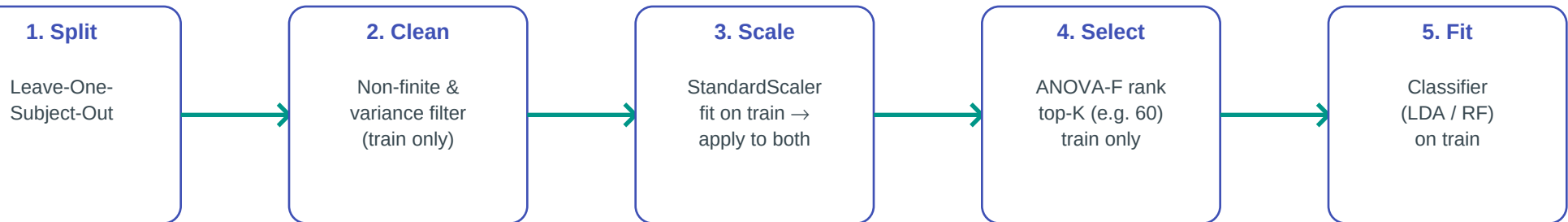
Model of choice

LDA + shrinkage

Parsimonious, stable,
interpretable; ensembles
comparable but overfit

Take-away for practitioners

- Multiband + multifeature extraction yields substantially higher AUC than any single-band approach.
- CZ nonlinear measures dominate single-feature rankings — a compact 5–10 feature LDA is often enough.
- Always validate with LOSO; within-subject CV will overstate performance on this problem.
- Keep a per-subject QC gate before any downstream decision or deployment.



6. Evaluate & Aggregate

- Predict probabilities on the held-out subject
- Collect per-fold AUC, sensitivity, specificity, confusion matrix
- Report mean AUC $\pm$ SD across subjects; flag chance-level subjects
- Ablations: paired Wilcoxon signed-rank on the same fold splits

■ **Never fit scaler, variance threshold or feature selector on the full dataset.**

Research / Exploration

- Random Forest / Gradient Boosting
- Useful for feature importance ranking
- Watch for overfitting on small N
- Use same LOSO folds for fair comparison
- Inspect per-subject performance variance

Deployment / Interpretability

- LDA with shrinkage estimation
- Top 5–15 CZ nonlinear features
- Fast, stable, low-latency inference
- Easy to audit & explain clinically
- Add per-subject QC gate first

Practical tip

A compact LDA on CZ Higuchi / DFA / SampEn / Petrosian / low-gamma often matches or approaches full 135-feature performance while remaining transparent and suitable for real-time streaming.

Always re-run LOSO after any feature-set change.

Baseline (≈ 135 features) vs. Baseline + family

- 1 Use identical LOSO subject splits for every comparison
- 2 Add one family at a time: Wavelet $\rightarrow$ MFCC $\rightarrow$ AR $\rightarrow$ Connectivity
- 3 Compute per-fold AUC difference (baseline+family – baseline)
- 4 Test with paired Wilcoxon signed-rank across folds
- 5 Record full config JSON + fold-level CSV for every run
- 6 Flag subjects that remain near chance after enrichment

Audit artifacts produced by 06_biomarker_loso_ablation.py

biomarker_loso_ablation_results.csv · biomarker_loso_ablation_folds.csv
biomarker_loso_ablation_config.json · per_subject_accuracy.csv

Feature matrices	Results tables	Ablation & config	Figures (examples)
• all_subjects_features.csv • features_xy.npz (X, y, subjects, feature_names) • biomarker_exploration_features.*	• classification_results.csv • per_subject_accuracy.csv • subject_summary.csv • biomarker_exploration_screening.csv	• biomarker_loso_ablation_results.csv • biomarker_loso_ablation_folds.csv • biomarker_loso_ablation_config.json • experiment hash / seed recorded	• results_panel.png • feature_heatmap.png • erp_analysis.png • biomarker_exploration_panel.png

- ☐ 1. Confirm channel naming; create channels.tsv mapping to 10–20 labels
- ☐ 2. Set target sampling rate 256 Hz, notch 50/100 (or 60/120), bandpass 0.5–80 Hz
- ☐ 3. Define pain & baseline epoch windows relative to stimulus onset
- ☐ 4. Run amplitude / SNR artifact rejection; save per-epoch QC metrics
- ☐ 5. Extract the baseline 135-feature set; store X, y, subjects, feature_names in NPZ
- ☐ 6. Execute LOSO with train-only scaling + ANOVA-F top-K selection
- ☐ 7. Run family-wise ablations (same folds) and Wilcoxon tests
- ☐ 8. Produce ROC, heatmap, per-subject accuracy plots + summary table
- ☐ 9. If stable biomarkers appear, retrain compact LDA and gate by subject QC
- ☐ 10. Validate on an external cohort when available (gold-standard step)

Aditya/

EDA scripts, feature-extraction code, PNG figures, lep_feature_extraction.py

Amar/

Core notebooks 01–05, 06_biomarker_loso_ablation.py, preprocessed/ outputs, requirements.txt

Hrishi/

Feature-importance notebook and supporting analysis

docs/

pipeline_slides.md, pipeline_slides_v2.md, generated PPTX/PDF

scripts/ & root

make_presentation.py, generate_pdf_slides.py, PDF_GENERATION_GUIDE.md

Self-descriptive names keep matrices auditable

CZ_higuchi	→ Higuchi fractal dimension at vertex
C3_alpha_power	→ Absolute alpha power at C3
CZ_C4_lowgamma_plv	→ Phase-locking value low- γ between CZ–C4
C3_MFCC3_mean	→ 3rd MFCC mean coefficient at C3
CZ_dfa	→ Detrended fluctuation analysis exponent

Recommended storage pattern

- CSV: rows = epochs, columns = features (human-readable audit)
- NPZ: X ($n_{\text{epochs}} \times n_{\text{features}}$), y, subjects array, feature_names list
- Always keep a matching subject_summary.csv and per-epoch QC file
- Hash the exact preprocessor + extractor config into a JSON sidecar

Version everything

Commit feature matrices, config JSON, fold CSVs and random seeds for every experiment.

Train-only stats

Scaler, variance threshold and ANOVA ranking must be fit inside each LOSO loop.

QC gating

Reject or flag subjects whose data quality falls below pre-defined SNR / rejection-rate thresholds.

External validation

When possible, test the final compact model on an independent cohort or different lab's data.

Real-time path

Once stable, port the LDA coefficients + ROI filters to a streaming inference service.

Documentation

Keep the README checklist and this slide deck updated with any protocol change.

Data & Code

Dataset: OpenNeuro ds005284 (Zhao et al.) — CC0 license

Core pipeline & LOSO scripts: Amar/ notebooks + 06_biomarker_loso_ablation.py

EDA & feature extraction: Aditya/

Feature importance exploration: Hrishi/

Repository: github.com/AmDeep/EEG-Analysis-Review

This deck synthesises the README, pipeline_slides.md and the executable generation scripts already present in the repo.



Thank you

Questions & next steps welcome

github.com/AmDeep/EEG-Analysis-Review

OpenNeuro ds005284 · Leakage-safe LOSO · Deployable LDA